

# Generative IA — Final Project Guidelines

**Course:** Generative AI

**Institution:** IE University

**Deliverable:** Functional MVP & Final Presentation

## 1. Project Overview

The final project for this course requires you to develop a fully functional **Minimum Viable Product (MVP)** that addresses a real-world business problem using GenAI or Agentic solutions. You will work in your assigned groups to build both the analytical backend and a user-facing frontend, demonstrating how advanced GenAI implementations can drive tangible business value.

## 2. Core Requirements

### Backend (The AI Model)

Your backend must be powered by a GenAI or Agentic architecture.

### Frontend & Integration

- **Frontend:** You must build an interface that allows a non-technical user to interact with your model. You are fully permitted—and encouraged—to use **vibe coding** (leveraging AI-assisted tools like Cursor, v0, Bolt.new, or Streamlit) to build this quickly.
- **Integration:** A critical part of the grade is the seamless end-to-end integration. The frontend must successfully pass user inputs to your generative ai backend and display the results in real-time.

## 3. Evaluation Criteria

Your project will be graded based on three core pillars:

### **Technical Depth & Execution**

- Proper data preprocessing, feature engineering, and handling of datasets.
- Justification of the chosen framework
- Robustness of the frontend-backend integration.

### **Business Use Case & Viability**

- Clear identification of a compelling, realistic business problem.
- A well-defined value proposition: How exactly does this predictive model save time, reduce costs, or increase revenue?
- Usability of the MVP—how intuitively the frontend solves the user's pain point.

## Presentation & Delivery

- Clarity, professionalism, and structure of the final pitch.
- Ability to translate complex deep learning mechanics into executive-level business insights.
- Quality and smooth execution of the live MVP demo.

## 4. Final Deliverables & Submission

You are only required to submit two assets via the campus platform:

1. **GitHub Repository Link:** Must contain clean, documented code for both the backend/model training and the frontend application, along with a brief [README.md](#) file on how to run the project.
2. **Final Presentation Deck:** Your presentation slides (PDF format preferred).

## 5. Presentation Day Rules

- **Time Limit:** Exactly **15 minutes per group**. This includes your pitch, the live MVP demo, and a brief Q&A session. Time limits will be strictly enforced.
- **Participation:** **All members of the group must actively speak** during the presentation.

| Criteria | Exceptional<br>(90-100%) | Proficient<br>(80-89%) | Developin<br>g<br>(70-79%) | Unacceptable<br>(<70%) | Weig<br>ht |
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|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Business Use Case &amp; Value Proposition</b> | The business problem is highly compelling and realistic. The value proposition is perfectly defined with clear impact metrics (ROI, cost reduction, efficiency).                                                                                              | Problem is clear and realistic. The model addresses it well, though business impact metrics could be more precise or quantified.                                                                                           | The business case is vague or unrealistic. Weak connection between the model's output and actual business value.                                                                                                  | No clear business problem identified, or the proposed solution does not match the problem at all.                                                                                   | <b>20%</b> |
| <b>Technical Depth &amp; Model Architecture</b>  | Flawless prompt engineering pipeline, embedding strategy, or fine-tuning approach. Strong architectural justification for the selected GenAI framework (e.g., RAG vs. Fine-Tuning) and agent design (e.g., single-agent, multi-agent orchestrations, tool-use | Sound model choice and implementation. Good system evaluation, but lacks deep justification for specific orchestration frameworks, hyperparameter/temperature settings, vector database configurations, or specific prompt | Solution runs but the choice of LLMs or agentic architecture is poorly justified. Weak evaluation metrics, inadequate guardrails, or clear signs of unaddressed agent loops, infinite reasoning cycles, or severe | Broken code, inappropriate model/framework choice for the business or technical problem, or fundamental misunderstandings of Generative AI, prompt workflows, or agentic paradigms. | <b>25%</b> |

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|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------|
|                                                 | <p>loops).</p> <p>Excellent evaluation metrics (e.g., LLM-as-a-judge, RAGAS, correctness, latency) and robust guardrails/mitigation for hallucination, security risks, or agent loops.</p> | <p>optimization steps.</p>                                                                                                          | <p>hallucinations..</p>                                                                                                               |                                                                                                                 |                   |
| <p><b>MVP Integration &amp; Frontend UX</b></p> | <p>Seamless frontend-backend integration. The interface (built via vibe coding) is highly intuitive, professional, and flawlessly handles real-time predictions.</p>                       | <p>Successful end-to-end integration. The interface works well, though it may have minor UX friction or lag during predictions.</p> | <p>Integration is fragile or partially hardcoded . The frontend exists but struggles to communicate dynamically with the backend.</p> | <p>Frontend and backend are completely disconnected. The model cannot be interacted with via the interface.</p> | <p><b>25%</b></p> |

|                                         |                                                                                                                                   |                                                                                                                                  |                                                                                                                          |                                                                                                                   |            |
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| <b>Presentation &amp; Team Delivery</b> | Exceptionally professional, executive-level pitch. Seamless transitions between <b>all</b> speakers. Highly engaging visual deck. | Clear and professional presentation. All members spoke, though transitions may have felt slightly disjointed. Good quality deck. | Disorganized flow. Distribution of speaking time was highly unequal among members. Slides are cluttered or hard to read. | Unprofessional delivery. Not all members spoke. Reading directly from notes/slides without engaging the audience. | <b>20%</b> |
| <b>Live Demo &amp; Time Management</b>  | Live demo worked perfectly under pressure. Presentation concluded exactly within the <b>15-minute limit</b> with time for Q&A.    | Live demo was successful despite minor hiccups. Finished slightly over/under time but managed content well.                      | Demo failed or was replaced by static screenshots. Went significantly over time, forcing the professor to cut them off.  | No demo attempted. Utter disregard for the time limit or presentation structure.                                  | <b>10%</b> |